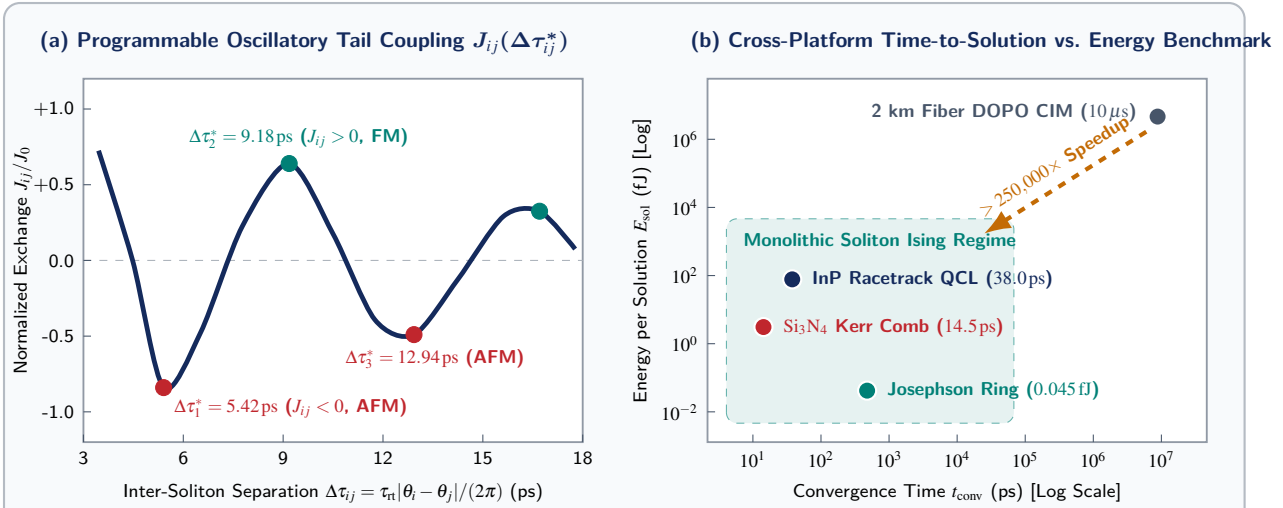
(a) Programmable Oscillatory Tail Coupling $J_{ij}(\Delta\tau_{ij}^*)$
+1.0
+0.5
0.0
-0.5
-1.0
Normalized Exchange J_{ij}/J_0
$\Delta\tau_2^* = 9.18\,\text{ps}$ ($J_{ij} > 0$, FM)
$\Delta\tau_3^* = 12.94\,\text{ps}$ (AFM)
$\Delta\tau_1^* = 5.42\,\text{ps}$ ($J_{ij} < 0$, AFM)
3
6
9
12
15
18
Inter-Soliton Separation $\Delta\tau_{ij} = \tau_R|\theta_i - \theta_j|/(2\pi)$ (ps)
(b) Cross-Platform Time-to-Solution vs. Energy Benchmark
2 km Fiber DOPO CIM (10 μs)
$\sim 250{,}000\times$ Speedup
Monolithic Soliton Ising Regime
InP Racetrack QCL (38.0 ps)
Si_3N_4 Kerr Comb (14.5 ps)
Josephson Ring (0.045 fJ)
10^6
10^4
10^2
10^0
10^{-2}
Energy per Solution E_{sol} (fJ) [Log]
10^1
10^2
10^3
10^4
10^5
10^6
10^7
Convergence Time t_{conv} (ps) [Log Scale]